Small Language Models for Shallow Knowledge Extraction

Milestone 2: Exploratory Data Analysis

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1. Introduction

This milestone presents an exploratory data analysis (EDA) of the BPDP dataset for the project Small Language Models for Shallow Knowledge Extraction. The objective is to examine the dataset structure, RDF graph characteristics, class distribution, and data quality before further experiments.

2. Dataset Description and Setup

The BPDP dataset contains 406 RDF files divided into Training and Testing subsets.

- Training: 100 True and 100 False files.
- Testing: 103 True and 103 False files.

The dataset contains 203 files in each class, providing a balanced distribution. Each RDF file represents information through subject-predicate-object triples.

The analysis was performed using Python 3.12.10 in Jupyter notebooks within Visual Studio Code. RDFLib was used to parse RDF files, pandas and NumPy to organise statistics, and Matplotlib to generate visualisations.

3. Analysis Methodology

All 406 files were examined in four groups: Training True, Training False, Testing True, and Testing False. The analysis included:

- Counting files and RDF triples.
- Measuring graph sizes and predicate frequencies.
- Examining unique subjects, objects, and object types.
- Inspecting and comparing selected True and False graphs.
- Checking date literals for possible formatting issues.

4. Results

RDF graph statistics. The dataset contains 8,000 triples. Each training category contains 1,976 triples, averaging 19.76 per file. Each testing category contains 2,024 triples, averaging approximately 19.65 per file.

The graphs contain 14, 17, or 20 triples. Most graphs (94 in each group) contain 20 triples. The True and False groups have identical graph-size distributions within each split.

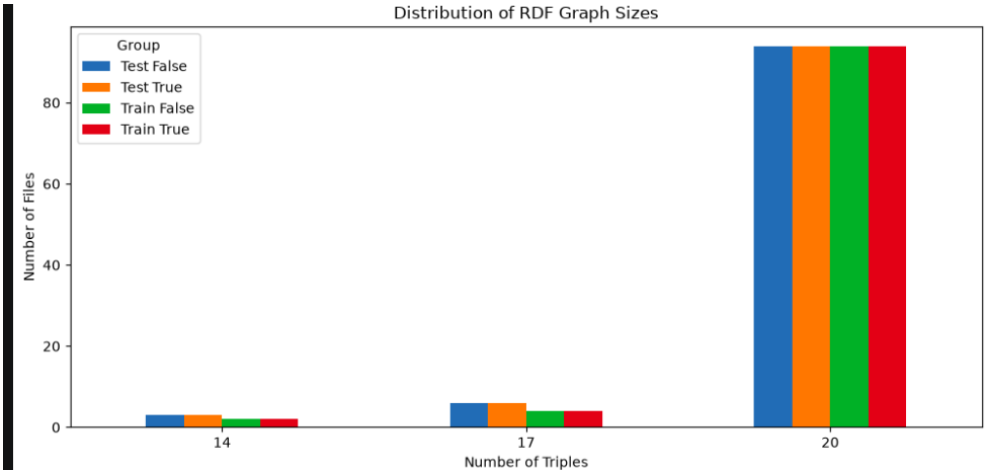


Figure 1: Distribution of RDF graph sizes across the four dataset groups.

Predicate frequency. Nine unique predicates were identified in each group. These include `rdfs:label`, `skos:altLabel`, `owl:sameAs`, and DBpedia predicates related to birth, death, and dates. Their frequency distributions are highly similar across the four groups.

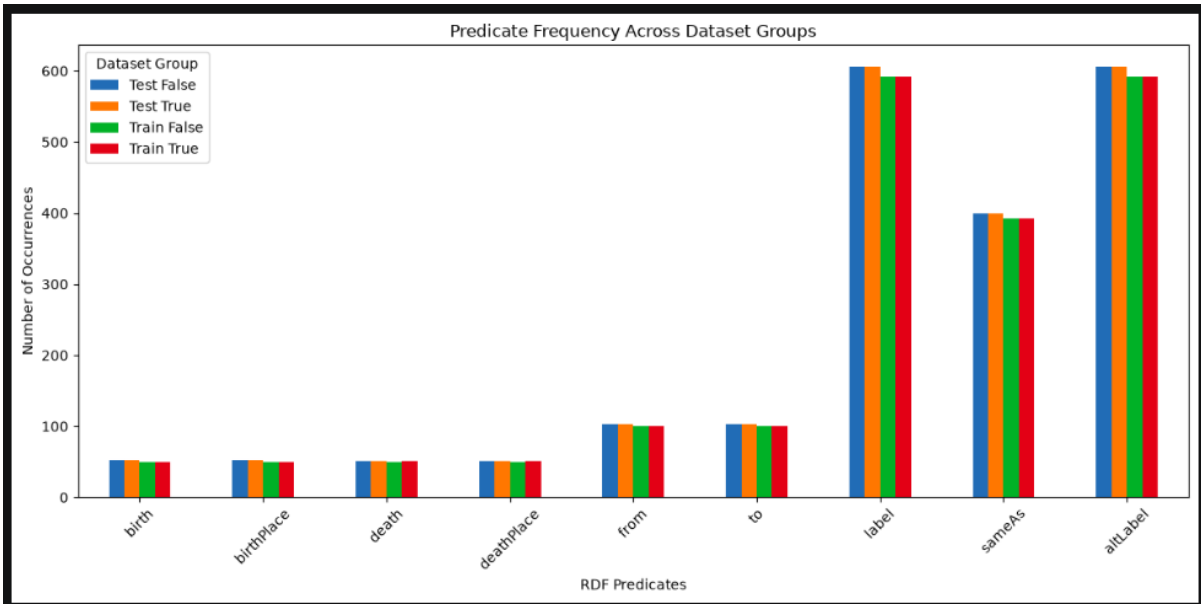


Figure 2: Predicate frequency across the four dataset groups.

The charts show similar graph-size distributions and predicate frequencies between the True and False categories. These structural statistics alone are unlikely to distinguish the two classes.

Subject and object analysis. Each training group contains 175 unique subjects and 779 unique objects. Each testing group contains 188 unique subjects and 846 unique objects. Literal objects are more frequent than URI objects, and no blank-node objects were identified. The aggregate statistics are identical between the True and False groups within each split.

True and False graph comparison. Two example training graphs contained 20 triples each, with 11 shared triples and 9 unique triples per graph. The main factual difference concerned George Meade's birthplace: Cádiz in the True example and Philadelphia in the False example. The remaining differences mainly described the respective locations.

A further comparison of ten graphs from each training category showed varying amounts of shared information. Files were paired by sorted order rather than verified correspondence, so this comparison is exploratory rather than a formal class-level evaluation.

5. Data Quality and Challenges

Python environment issue. An initial pandas import failed because of a DLL loading problem. The environment was configured to use Python 3.12.10, and the required libraries were installed successfully. The imports and analysis then executed.

Date-format warnings. RDFLib generated conversion warnings for some literals declared as `xsd:date`. For example, `1846-4-4` does not follow the expected zero-padded format `YYYY-MM-DD`. A preliminary regular-expression scan identified 644 possible formatting issues across 322 files.

The original dataset was not modified. The warnings did not prevent the RDF graphs from being parsed for the reported statistics. The scan was preliminary and did not validate every value as a calendar date.

6. Conclusion

The EDA identified a balanced dataset with similar aggregate RDF structures across the True and False categories. Graph sizes, predicate frequencies, and subject/object statistics alone do not reveal a clear distinction between the classes. However, individual graph comparisons showed that different factual information can be represented by graphs with similar structures.

Potential date-format inconsistencies were also identified for consideration during future preprocessing. These findings provide a basis for the next stages of the project.